# Electric Flux and Gauss's Law

Count the lines through a closed surface, and the charge inside tells you how many there are

> SYNOPSIS
>
> Back in **Note 3** a phrase went past almost in passing — **flux is line density**. This note collects on it. Draw any closed surface you like around some charge, count the field lines crossing out of it, and that count depends on **nothing but the charge inside** — not the shape of the surface, not its size, not where the charge sits within it. That is Gauss's law, and where a problem has enough symmetry it hands you the field in two lines of algebra where the direct integral takes two pages. We will do it the hard way first, so that you can see what is being saved.

## 1. Electric Flux

Flux is the simplest idea in this note and it is worth stating in his own shorthand before any integral signs appear:

$$\Phi_E = \{\text{electric field}\} \times \{\text{area}\}$$

But *which* area, and how much of the field? Take the same patch of area twice and put it in two different places:

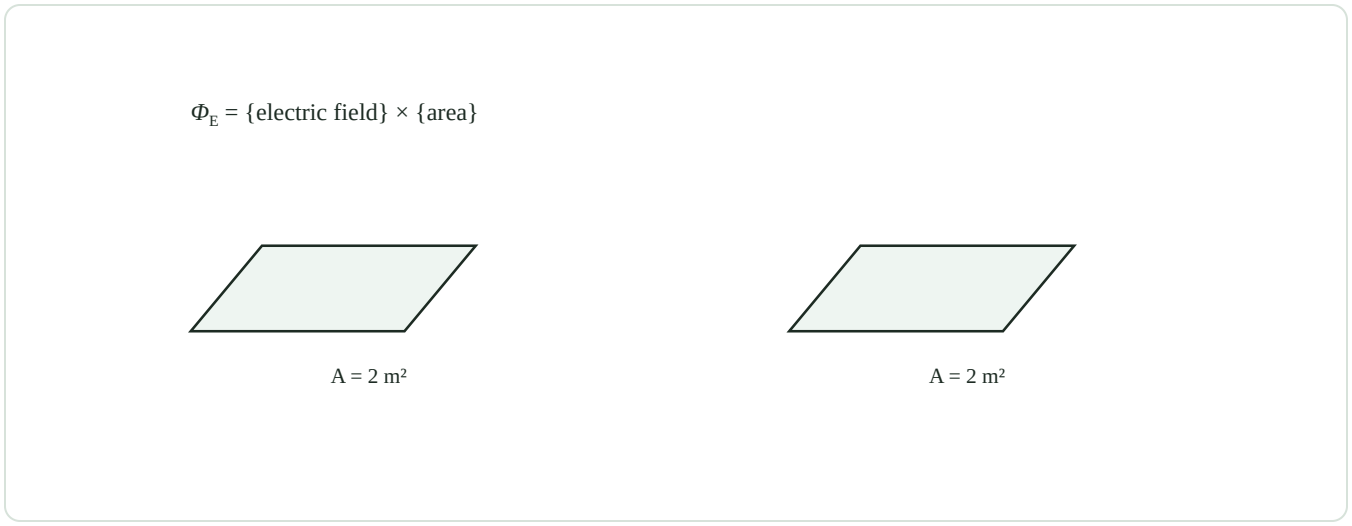



**Figure 1.** Draw the field lines through each patch — many on the left, few on the right — then say in one line why the flux differs when the area does not.

**Your job on Figures 1 and 4.** The patches are drawn for you and the field is not. On **Figure 1** draw the field lines through each patch — many on the left, few on the right — and say why the flux differs when the area does not. On **Figure 4** draw **E** leaning away from the normal and project it onto $\hat{n}$.

Now close the surface up, so it wraps the charge entirely. Two things can happen, and he labels both:

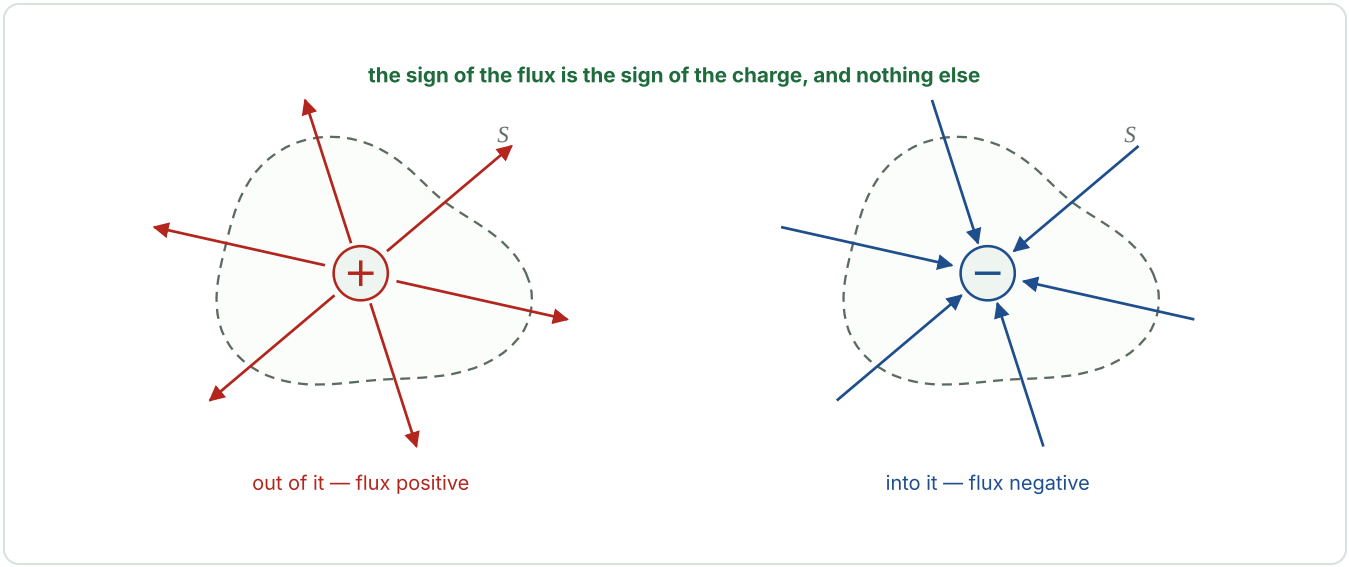



**Figure 2.** Close the surface up so it wraps the charge entirely, and two things can happen. Lines out: **flux positive**. Lines in: **flux negative**. **The sign of the flux is the sign of the charge, and nothing else.**

"Closed" is doing real work in that sentence, and his page tests it before going on:

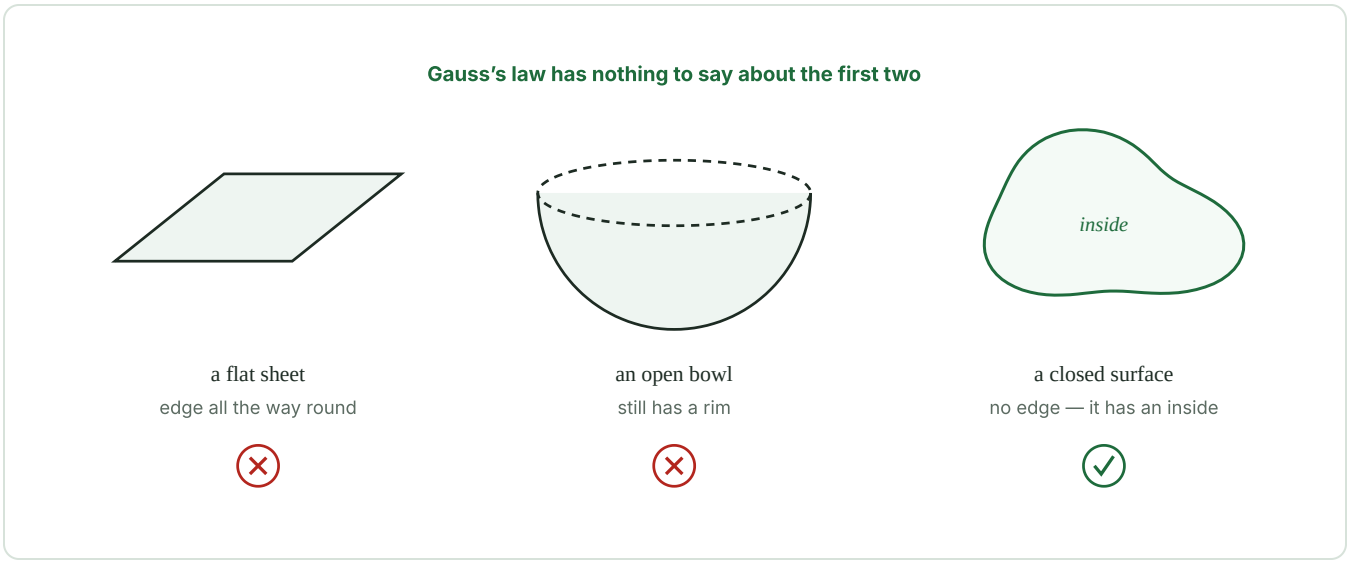
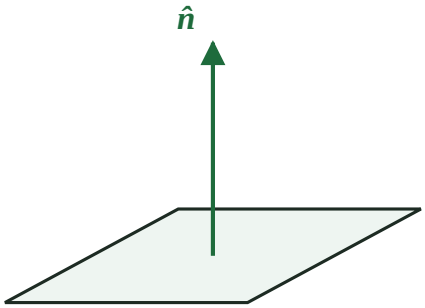



**Figure 3.** "Closed" is doing real work in that sentence. The first two surfaces have an edge you could walk off; only the third separates an inside from an outside. **Gauss's law has nothing to say about the first two.**

The one subtlety is that a field skimming *along* a surface carries nothing through it. Only the part along the outward normal counts, which is what the dot product is doing in every formula below:




$d\Phi = \mathbf{E} \cdot \hat{n}\, dA$

**Figure 4.** Draw **E** leaning away from the normal, then project it onto $\hat{n}$. What happens to the flux when **E** lies flat in the surface?

Put those together and you have the law, in the two forms he writes it:

$$\boldsymbol{\varepsilon_0 \Phi = q_{\text{enclosed}}} \quad \text{for a closed surface!!}$$

$$\Phi = \oint_S \mathbf{E} \cdot \hat{n}\, dA = \oint_S \mathbf{E} \cdot d\mathbf{A} = q_{\text{enc}} / \varepsilon_0$$

**WHAT WE ARE INTERESTED IN, IN HIS WORDS**

$$\Phi = \oint_S \mathbf{E} \cdot d\mathbf{A} = q_{\text{enc}} / \varepsilon_0$$

He boxes this one and writes **“what we are interested!!”** beside it, with **“Gauss’ Law”** underneath. Everything else in the note is this equation being used.

## 2. Solving the Electric Field with Gauss’s Law

Before trusting it on anything hard, check it on something whose answer is already known. His note at the top of this page: **“We know this is solution.”**

$$\mathbf{E} = (1 / 4\pi\varepsilon_0)\,(1 / r^2)\ \hat{r}$$

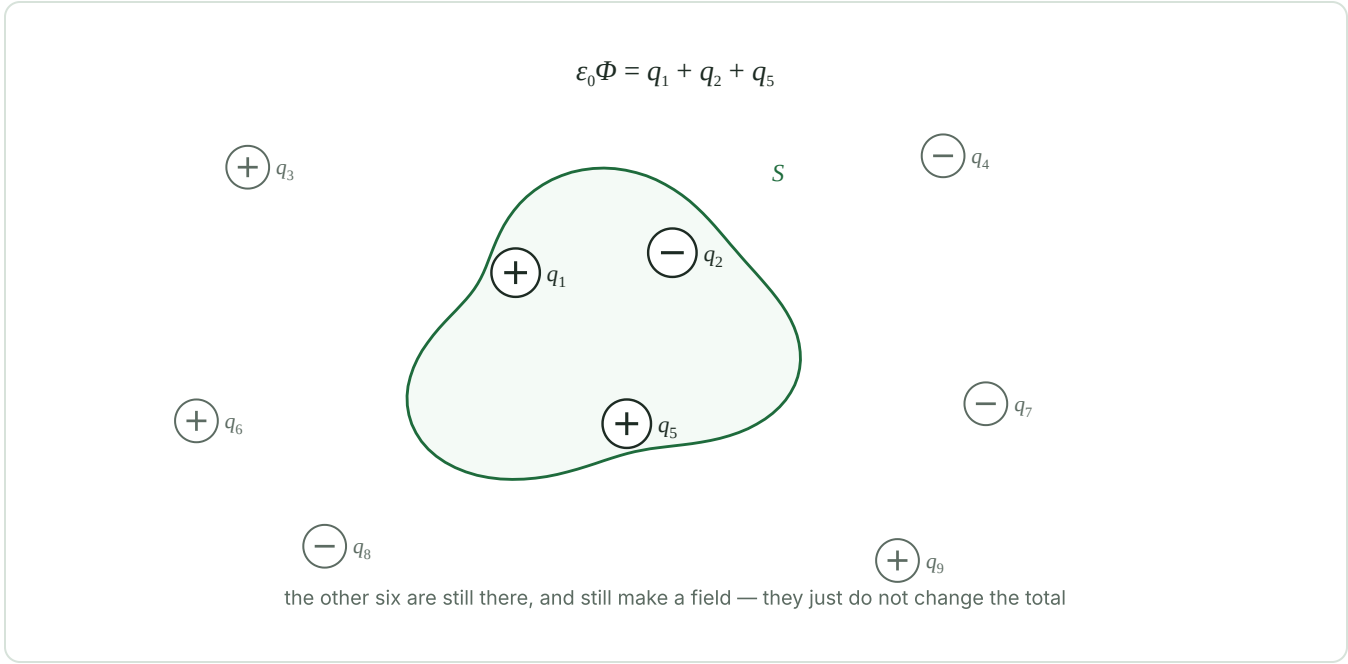



**Figure 5. The whole content of the law in one picture.** Nine charges, three of them inside, and only those three appear in the flux. The other six **still make a field everywhere on the surface** — what they do not do is change the total through it.

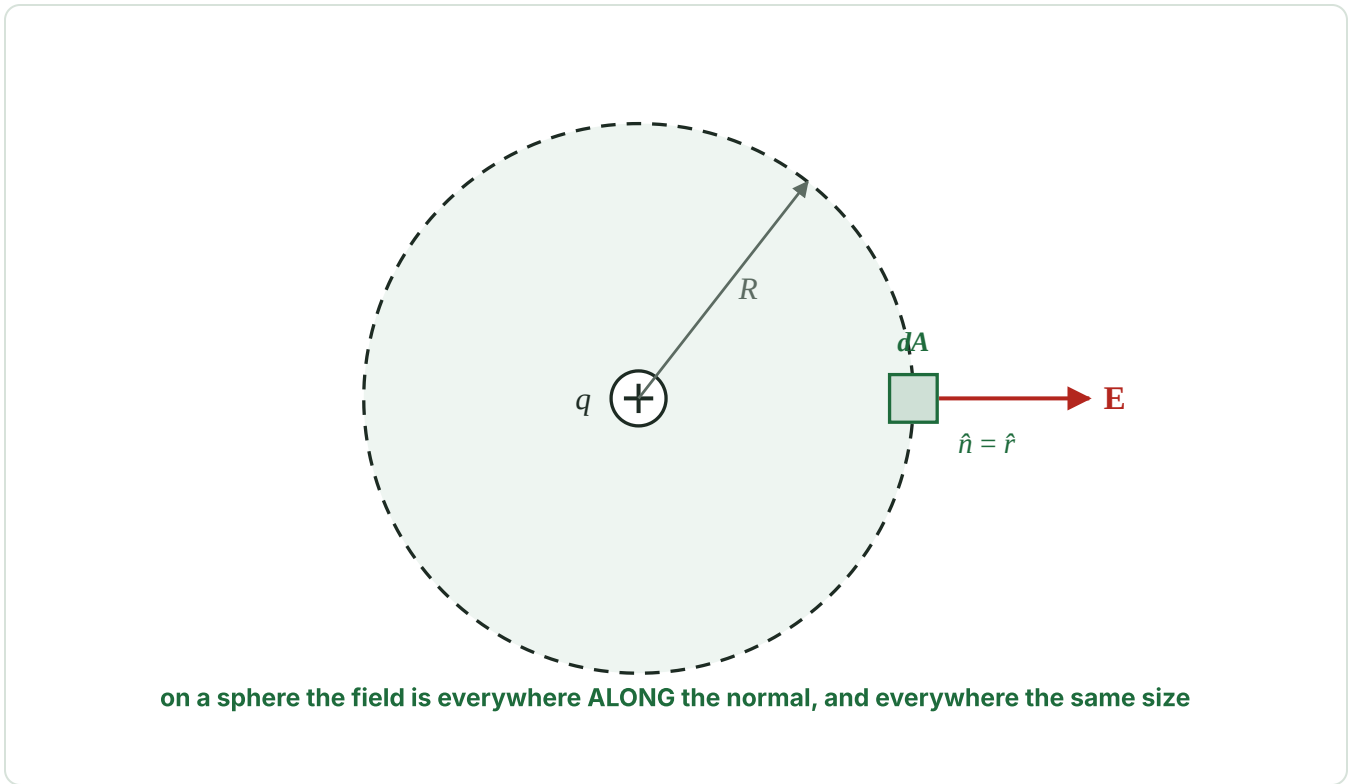



**Figure 6.** A point charge at the centre of a spherical surface. Two things make this work, and they are the two to look for in every problem below: $\hat{n} = \hat{r}$ everywhere, so the dot product is just a product; and ***E* is the**

**same size everywhere on the surface**, so it comes out of the integral.

His heading for what follows is **“Demonstrate Gauss Law”**:

He then works the law on a set of cases where nothing has to be integrated at all — the whole job is deciding which charges are inside:

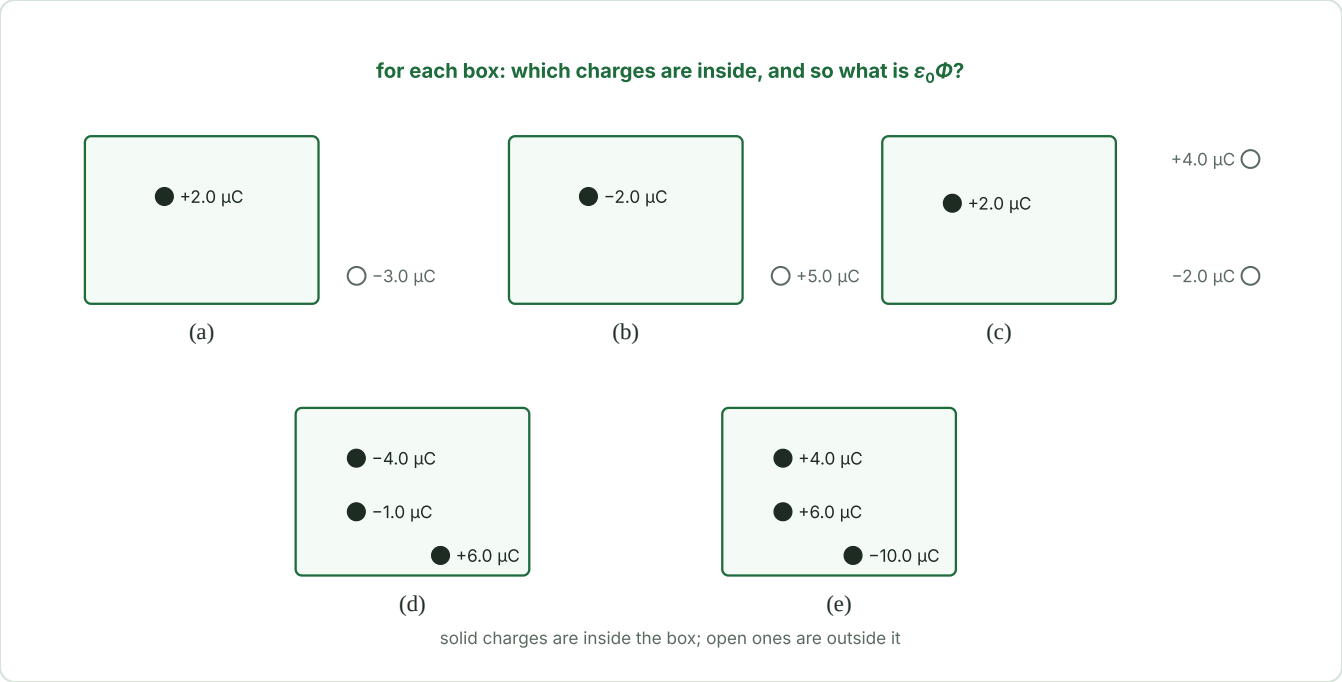



**Figure 7.** Five surfaces, and the whole job is deciding which charges are inside. **(e)** is the one worth pausing on: +4, +6 and −10 all sit inside, and they cancel.

**ONE TO CHECK FOR YOURSELF**

Work out $\varepsilon_0\Phi$ for all five boxes in **Figure 7**, and give every answer in **μC**. **Watch the exponent.** A charge quoted in μC is $10^{-6}$ C, and that factor is the easiest thing on this page to drop on the way through. **(c)** is the one to do most carefully.

**WHERE THIS IS GOING**

$$\varepsilon_0\Phi = q_{\text{enclosed}}$$

That is the law, and it is now proved on the one case whose answer was already known. **Note 7** turns it loose on cases whose answers are not — and, so that you can see what it is saving you, does three of them the long way first.

## Looking ahead

**Note 7** takes three charged objects — a line, a ring and a disc — and finds the field of each by direct integration, which is honest work and takes pages. Then it points Gauss's law at the same objects. **Two of the three come back in two or three lines. One does not**, and that one marks the boundary of the method.